This page lists the thirteen nodes with their owner and the ledger events each writes.

NODE	OWNER	WHAT IT DOES	DISTINGUISHING LEDGER EVENTS
`ingest_events`	rules	Reads the pack's structured events into the twin.	`event_ingested`
`classify`	rules	Counts the structured events by type and routes to fusion when an advisory is present.	`rule_eval`
`fuse_advisory`	language model	Schema-constrained adaptive vote, three samples escalating to five; keys allow-listed; output carries UNTRUSTED_FREETEXT.	`llm_call · model_rationale`
`fusion_gate`	rule	Refuses to act below completeness 0.60; consults the shift memory of the source, which can add a person and cannot remove one; ingests the fact.	`rule_eval · policy_gate · tool_call`
`assess_feasibility`	twin, CP-SAT	Recomputes margin at the P90 of the transfer distribution; with several connections at risk, calls replan_terminal for one joint plan against the live shared budgets.	`tool_call`
`plan_options`	rules	Enumerates the single-connection options; a dissent check re-derives the margin from the contract formula without the option generator.	`tool_call`
`policy_gate`	rules	Table lookup by action class and arguments; row 10 denies an unlisted class; a deterministic scope check on the arguments follows.	`policy_gate`
`request_approval`	duty officer	Interrupts with the card; the approval server mints a single-use token on APPROVED; an edit re-runs the gate and re-binds the token.	`approval_requested · approval_granted · approval_denied · approval_timeout_deny`
`execute_actions`	gated write	Presents the token to portnet-mock; the gate checks token, credential, idempotency key and degraded state before the fault layer.	`action_executed · action_failed`
`verify_effect`	twin	Re-runs feasibility after the write and records the new margin.	`tool_call`
`close_episode`	rules	Records the outcome to shift memory, emits the handover note, loops to triage while steps remain.	`replay_marker`
`escalate`	rules	Writes the escalation summary for the duty supervisor, naming each unsaved connection.	`escalated · replay_marker`
`degrade_monitor`	rules	Flips the episode to DEGRADED_TO_ADVISORY while a degrading fault is active and records the clearance.	`degraded_mode_entered · recovered`

rule_eval and tool_call also appear on most rows and are listed only where they distinguish the node.

TOOL SERVERS, ONE FROZEN CONTRACT (SELECTED TOOLS)

SERVER	TOOLS
`twin-mcp`	`feasibility_check · replan_options · replan_terminal · simulate_what_if · ingest_fact · ingest_event · get_connections`
`portnet-mock-mcp`	`get_vessel_schedule · get_box_group · get_yard_state (reads) · set_transfer_priority · request_cutoff_extension · propose_rebooking · create_restow_order (gated writes).` Named for PORTNET in the code; at PSA the yard writes belong to the terminal operating system and the community-facing requests to PORTNET.
`approval server`	`request_card · wait_decision · decide · get_card · verify_token; decide is the only path that mints a token`
`fault injector`	`inject · clear; A2A_TIMEOUT, AGENT_MISROUTE, APPROVER_UNREACHABLE, CONTEXT_OVERFLOW, CORRUPTION, GUARDRAIL_BYPASS, INFINITE_LOOP, LATENCY, TOOL_FAILURE, WRONG_TOOL`

TIER ROUTING

Three tiers: rules; local, which is llama3.2:3b through Ollama with tokens measured and cost imputed at $0; and frontier, which is off by default, keyed by environment variable and priced at a dated list price. Routing is rule-based, the frontier fires on defined triggers such as low vote agreement or a detected contradiction, a kill switch overrides the routing, and per-tier hit counters are printed on each run.

LEDGER

this_hash is the SHA-256 over the canonical JSON of the event without this_hash, and the first prev_hash is 64 zeros. A head anchor over (count, tip) makes truncation visible. Each event carries the CSA 4.3 field set: action, digests of inputs and outputs, state change, error, timestamps, correlation id, acting credential, tier, tokens. Editing any field of a past event changes every hash after it, so the edit is visible on replay.

Each row is a lookup by action class, and the risk basis is severity, reversibility and the feasibility of oversight.

#	ACTION CLASS (WRITE TOOL)	TIER, AS THE CODE ENFORCES IT	BUDGET
1	Read or query terminal state (twin reads, portnet.get_*)	T2, open class	**60 calls/min**
2	Risk annotation, internal operations notification	T2	**20/shift**
3	Expedite yard transfer (portnet.set_transfer_priority)	T1	**5/shift**
4	CRITICAL transfer priority, preempts other cargo	T1; HIGH risk, written justification required	**2/shift**
5	Cut-off extension request (portnet.request_cutoff_extension)	T1; HIGH risk, written justification required	**3/shift**
6	Rebooking proposal (portnet.propose_rebooking)	T1; HIGH risk, written justification required	**3/shift**
7	Restow order (portnet.create_restow_order)	T1; HIGH risk, written justification required	**2/shift**
8	Escalation summary to the duty supervisor	T2	**10/shift**
9	Berth or berthing-time change (no write tool exists)	T0 only	**n/a**
10	Any action class not in this table	AUTO-DENY and escalate	**n/a**
11	Twin state ingest (twin.ingest_fact, twin.ingest_event)	T2	**120/shift**
12	Any option whose expected value is below its cost	ADVISE_ONLY, never proposed as a write; three numbers attached	**n/a**

The contract prose gives row 3 a T2 tier for repeats inside an approved plan and row 7 a HIGH risk only when a dangerous-goods class is set; the code applies the stricter reading in both rows, which is what the table above prints. Budgets are demo placeholders held as one counter per action class terminal-wide; production scopes each per terminal, berth or service, which is configuration of the same table. The sixth expedite in a shift is refused RATE_LIMITED and lands as action_failed; inside a joint plan the remainder is re-allocated under the budget that is left, and otherwise the episode escalates with a written summary.

MAPPING TO PUBLISHED FRAMEWORKS

RELAY TIER	IMDA MGF V1.5 HUMAN INVOLVEMENT	CSA LEVEL
T0 advise	Agent proposes, human operates	L0 to L1
T1 ask and approve, editable plan	Agent operates, human approves; agent and human collaborate	L2
T2 act and audit	Agent operates, human observes, post-hoc audit	L2
not built		L3, so every path stays enumerable

T2 sits at L2 with T1 because its action classes are reads, annotations and summaries that change no terminal state; the mapping is an alignment claim, and no certification has been sought.

DENY BY DEFAULT, ENFORCED

The 120-second window is a wall-clock timer enforced on the server. A card the console shortens for recording is labelled with the shortened window and the contract default. Readiness is computed from the predicates the refusing layers apply, so the Approve button is disabled with the reason when the click could not land; the write gate stays the only control.

THE GOVERNED EDIT

An edit must resolve to a solver-enumerated option or a transfer-priority level; a free-form edit is refused, the card is denied and the episode escalates. The edited plan is re-simulated through the twin and must agree with the solver margin, the policy gate re-runs, and approval supersedes the original card with a new one whose argument digest re-binds the token. Trace entries inside the frozen event enum: a human_note labelled approval_card_edited, one tool_call labelled whatif_result per re-simulation, and a policy_gate re-run.

This page measures the language model's step on 200 advisories, then the live sweep, the shift memory and the external solver benchmark.

METRIC, N = 200 ADVISORIES	REGEX BASELINE	LLAMA3.2:3B, LOCAL	HYBRID ROUTER
Extraction accuracy	0.548	0.575	0.726
ETA invention rate, parse layer	0.110	0.477	0.119
Contradiction flag recall	0.471	1.000	1.000
Gate routing accuracy	0.865	0.845	0.850
False accepts end to end, of 200	4	10	4
Mean latency	0.0 s	37.6 s	40.3 s

The corpus is 64 canonical paraphrased advisories, 88 benign template variants and 48 adversarial items in six classes: prompt injection 12, fabrication bait 10, contradiction trap 8, malformed 8, unicode trick 8, oversized 2. On the adversarial subset the regex baseline is ahead of the local model on extraction and gate routing, and the hybrid router matches the baseline's 4 false accepts there. The 8B tier was not run on the recording machine.

The hybrid router compares the regex parse and the model parse field by field. Where the two disagree, or only one produced a date or time, it keeps the value only when, in the sentence where it appears, the nearest role marker is that role's; a tie or a missing marker drops the field. Of 126 ground-truth values present in their own text, 124 still ground; the two that do not belong to one advisory asking a person to clarify it, which escalates. The rule is lexical, so an adversary who writes arrival language directly before a cut-off value can still win the proximity contest; that buys one wrong field on an advisory that must still pass reconciliation, the gate, the policy table and a card.

EXTERNAL SOLVER BENCHMARK, PORT OF BARCELONA BERTH ALLOCATION

Ten published instances (quays 24B and 36A, 293 ships) solved with RELAY's pinned CP-SAT setup under a 120-second limit: 10 of 10 solved and verified by an independent feasibility checker, 8 proved optimal, 9 matching the published best-known solution and 1 improving it (bcn_36A_22, 179 against 180). This is the one result graded against an answer key this project did not produce; berth planning itself stays outside RELAY's write authority.

LIVE SWEEP, THE FULL GRAPH WITH A REAL MODEL, N = 100

Tokens per advisory episode, three-sample vote (retired build)	1,993
Latency per advisory episode, recording machine	17.9 s
Frontier counterfactual per episode, three-sample vote (retired build)	$0.0013
Episodes escalated of 100	11

LIVE COST PROFILE, FIVE-SAMPLE BUILD, 30 EPISODES

Tokens per advisory episode, current build	4,689.7

ADAPTIVE SAMPLING, WIRED

Baseline, full five-sample panel, tokens per advisory	4,555.8
Adaptive panel, tokens per advisory	3,017.6
Saving, every quality column unchanged	33.8%

Two rules force the full panel: a field agreed only as a rescaled ratio has not been agreed, and unanimity on a null forces the full panel where a deterministic text probe says the field should be present. Wiring it showed that the benchmark's quality columns could not see a dropped rotation change, and the two rules are the fix.

SHIFT MEMORY, REPLAYED ON THE 200 ROWS

False accepts without memory	10
False accepts with memory	6
Extra escalations introduced	1

Shift memory can route a source's facts to a person and cannot lift that requirement. It acts on recorded false accepts from that source, and a source being new has no effect.

Tokens measured off the ledger; dollars imputed at list price; no API was billed.

evalx/results/fusion-ladder.json · sweep-live-n100.final.json · cost-curve.json · efficiency-eval.json · memory-eval.json · external-benchmark.json

Each control was tested by fault injection, a red-team attack or a check that must fail on a tampered trace, and the certificate counts the results.

RED TEAM ON THE APPROVAL PATH, FOURTEEN ATTACKS

Threat model: the agent process is compromised, so the attacker calls any tool with any arguments, mints any string and replays anything seen, and cannot forge the approval server's pepper or write to its store.

ATTACK	FIRST TRY	NOW
No token, forged token, binding mismatch, denied card, timed-out card, expired token, cross-tool reuse, wrong credential, guardrail bypass, card rewritten after approval	refused	refused
A10, the agent approves its own card and spends the token	`write succeeded`	`MAKER_IS_CHECKER`
A12, one approved token spent again under a second idempotency key	`write succeeded`	`TOKEN_ALREADY_USED`
A13, twelve threads race the single-use spend	`writes succeeded`	`TOKEN_ALREADY_USED under a lock`
A14, a decided card re-registered and reset to PENDING	`write succeeded`	`CARD_ID_ALREADY_DECIDED`

WHAT EXECUTED IS WHAT THE OFFICER SAW, SEVEN CHECKS

D1 the chain verifies; D2 each executed action has an approval before it; D3 the approver is a human principal; D4 the executed argument digest equals the digest the card rendered; D5 the token was bound to that tool and digest; D6 an edited plan supersedes, with exactly one card approved; D7 an episode with no approval executed nothing. 5 of 5 episodes hold; six tests mutate a trace and require a breach, and two require a clean trace to pass.

SEEDED WRONG RECOMMENDATIONS

129 of 129 seeded wrong recommendations that fired were caught by the deterministic re-checks over 400 episodes, with 0 cards raised, 0 writes and 0 false flags on 101 unseeded control episodes. With the re-checks switched off, the same seeds were caught 0 times; that ablation is the figure that shows the re-checks do the work.

THE FALSIFICATION CERTIFICATE

Controls named by the deliverables, parsed from four documents	56
Probed: switched off, and a named green test went red	48
Named as unprobed, with the reason	5
Outside this code, listed by name	3
Probes run, caught, survived, invalid	`56, 56, 0, 0`

A probe counts as CAUGHT only when its watcher tests were green on the clean tree and, with the control off, a named test in a covering file failed. A missing, red, empty or unreachable watcher marks the probe INVALID, and a test feeds the probe script deliberately bad probes to require that. Two findings a green suite could not produce: deleting the static-root guard left thirteen server tests green because the HTTP client resolved the traversal before sending, so the guard's absence served application source; and a mutant restored within the same second was read from Python's bytecode cache, so probes now purge bytecode on both sides.

INDEPENDENT ORACLE, N = 320

A second implementation of the feasibility rule, written from the contract text: verdict agreement 1.00, at-risk population 188 against 188, rules-only catch 0.894 against it, false escalations 0 of 132. Seven single-point mutations of the rule were detected 4 of 7 times; the three undetected are the completeness gate moved from 0.60 to 0.50, the ETA weight moved from 0.30 to 0.20, and infeasibility counted only strictly below zero. The oracle certifies the margin arithmetic and leaves those three parameters to their own tests.

SECURITY REVIEW ROWS, SELECTED

S-0 no secrets in the tree; S-1 every write path behind one gate; S-2 token binding; S-3 cross-site request forgery, found and closed; S-9 single use at the approval server; S-12 the hash chain and replay refusal; S-16 the untrusted advisory channel; S-17 the approver edit path; S-20 the four red-team attacks that landed at first try; S-21 the head anchor against truncation; S-23 the grounding veto. The review runs to S-24, each row with a rerunnable command.

Scale profile 50 to 5,000 episodes with no super-linear growth; soak of 10,000 episodes with 0 failures.

`evalx/results/approval-attacks.json` · `oversight_chain.py` · `oversight_probes.py` · `mutation_probes.py` · `independent_oracle.py` · `docs/SECURITY-REVIEW.md`

This page lists the references, the results file behind each figure, and the four reproduction commands.

REFERENCES

PSA International, Annual Report 2025, Year in Review p.23: 17 services onboarded, 518 vessel calls, 84% achieving Assured Port Time targets. Sustainability Report 2025 p.4 (44.5 million TEU), p.33 and p.113 (Scope 3 745 ktCO2e; Category 9 at 47%, 347.2 ktCO2e).

IMDA, Model AI Governance Framework for Agentic AI, v1.5, May 2026: human-involvement levels, the editable plan, the four pre-deployment test dimensions.

Cyber Security Agency of Singapore, Addendum on Securing Agentic AI Systems, June 2026: per-agent identity (2.6), budgets and loop-breakers (3.1), trace fields (4.3), autonomy levels L0 to L3.

Kambhampati et al., "Position: LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks", ICML 2024. The agency boundary follows this pattern with the model's authority narrowed to facts.

Aggarwal et al., "Let's Sample Step by Step: Adaptive-Consistency", EMNLP 2023, which the fusion vote applies as a cheap panel that escalates.

Horvitz, "Principles of Mixed-Initiative User Interfaces", CHI 1999, the expected-utility threshold behind the pricing gate.

Sea-Intelligence, global schedule reliability 62.6%, June 2026. Kuehne+Nagel port operational update, Singapore yard density 80 to 85%, June 2026.

Santini, berth allocation problem instances, Port of Barcelona 2024, GPL-3, github.com/alberto-santini/berth-allocation-problems. The instances are downloaded at run time and are not committed to the repository.

RESULTS INDEX, ONE LINE PER FILE CITED IN THIS DECK

`ais-slip.json · ais-warning-lead.json`	silent slips, the slip window and the warning lead on the recorded days
`sweep-full-n500.final.json · sweep-full-n500-evgate.json`	500 seeded scenarios, oracle-gated, gate off and gate on
`save-value-audit-n500-evgate.json`	held-out re-pricing of every booked save
`validity-oracle-n320.json · oracle-mutation-power.json`	the independent feasibility oracle and its detection power
`fusion-ladder.json`	three-tier ladder on 200 advisories; injection resistance
`sweep-live-n100.final.json · cost-curve.json`	the full graph with llama3.2:3b; tokens and latency measured
`external-benchmark.json`	Port of Barcelona instances, 10 of 10
`approval-attacks.json · governance/results/attacks.json`	fourteen attacks on the approval path; twelve on the package
`mutation-probes.json · docs/CONTROL-CENSUS.json`	the falsification certificate inputs
`soak-profile.json · scale-profile.json`	10,000 episodes; the pre-fix append curve (post-fix figures in docs/SCALE-AND-VALIDITY.md)
`oversight-load.json · impact-model.json`	Erlang C and DES over 100,000 cards; 43 labelled inputs, both columns, tornado, bootstrap
`behaviours-conformance.json`	the six behaviours as checks over real ledgers

REPRODUCE, NO NETWORK, NO API KEY

```
.venv/bin/python -m stubs.selftest
.venv/bin/python -m pytest -q
.venv/bin/python evalx/scorecard.py
.venv/bin/python console/demo_walk.py
```

Code, tests and every result file are available to the judges on request.

Result files live under evalx/results/ unless another path is printed. The cascade episode with a local model: agentcore/replay.py --pack cascade.json --mode hybrid, about a minute with Ollama.

`evalx/results/ · docs/PRIOR-ART-AND-ORIGINALITY.md · README.md`